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Research Highlight

To be or not to be direct: The role of neuromedin U in neuro-eosinophil crosstalk

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With the development of new experimental techniques (e.g., single-cell RNA sequencing (scRNA-seq), light sheet microscopy, and clustered regularly interspaced palindromic repeats (CRISPR)/CRISPR-associated protein 9 (Cas9)), emerging studies have begun to uncover the interactions between the nervous system and the immune system, both in health and diseases. For example, in mucosal tissues such as the lung and the intestine, neurotransmitters including acetylcholine [1] and adrenaline [2] can directly regulate the immune cell functions thus affect the results of allergy, asthma, and anti-pathogen host defense [3].

The neuropeptide neuromedin U (NMU) also plays critical roles in regulating immune responses [3]. It is widely distributed and with the highest peripheral tissue expression in the gastrointestinal tract [4]. NMU has two receptors—neuromedin U receptor 1 (NMUR1) which is expressed in the peripheral tissues, and NMUR2 which is expressed in the central nervous system. Notably, NMUR1 was shown to be selectively expressed on group 2 innate lymphoid cells (ILC2s) among all immune cells [5,6]. In the lung and intestine, cholinergic neurons were detected in close proximity to ILC2s. A subset of cholinergic neurons are capable of producing NMU, which acts on the NMUR1 expressed on ILC2s to induce cell activation, proliferation, and type 2 cytokine (e.g., interleukin-5 (IL-5), IL-13) production [7], which in turn promotes the immune responses of eosinophils indirectly. However, NMU was shown to activate human peripheral blood eosinophils *in vitro* and NMUR1 also be detected in mouse eosinophil cell line Y-16 [8], suggesting that NMU has the potential to directly regulate eosinophil responses.

Eosinophils are produced in the bone marrow and reside in multiple tissues such as the gastrointestinal tract, lung, and adipose tissue. Notably, eosinophils that reside in the lamina propria of the small intestine exhibit distinct features, including prolonged lifespan [9] and signs of degranulation [10]. Although there were significant advances in the development of scRNA-seq, technical challenges still exist that impede transcriptomic analysis of eosinophils. The challenges include the shear stress during cell sorting and sample preparation, the RNase within eosinophils, and the degranulation of eosinophils—all of these result in transcript degradation and reduce the quality of the RNA library. As a consequence, the heterogeneity of eosinophils and the functional differences of their subgroups were largely unknown.

In a recent study, Li et al. [11] optimized the experimental workflow and techniques, and successfully collected eosinophils from multiple tissues in mice. First, they conducted bulk RNA-seq analysis on eosinophils isolated from seven different tissues of the mice—the bone marrow, blood, spleen, lung, skin, colon, and small intestine. Their results revealed tissue-specific molecular characteristics of eosinophils, which express different cytokines and receptors for pro-inflammatory signals in different tissues. Next, they performed scRNA-seq on small intestinal eosinophils to further characterize their transcriptional heterogeneity and revealed that the distinct transcriptional patterns of small intestinal eosinophils mainly attributed to the *Nmur1*-expressing subset of the cells.

This new finding from Li and colleagues [11] is inconsistent with previous studies [5,6], which report that *Nmur1* is solely expressed on ILC2s among all the small intestinal immune cells. This discrepancy may be caused by the BAC transgenic reporter mice used in the previous studies. It might not accurately reflect the expression level of endogenous NMUR1 [5,6]. Therefore, Li et al. [11] employed the CRISPR/Cas9 technology and generated a new knock-in mouse strain (*Nmur1*$^{\text{iCre-TdT}}$), which is more reliable in showing the endogenous NMUR1 expression. Using this new mouse strain, they confirmed the NMUR1 expression in small intestinal eosinophils. Consistent with their RNA-seq data, this reporter mouse strain also helped confirm that *Nmur1*-expressing eosinophils only presented in the small intestine but not in other tissues and organs. In addition, Li et al. [11] showed that NUMR1 is also expressed on human eosinophils isolated from intestinal biopsies.

Eosinophils play a critical role in host defense against parasitic infections [12]. However, excessive eosinophil responses (i.e., eosinophilia) are related to inflammatory diseases such as allergy,

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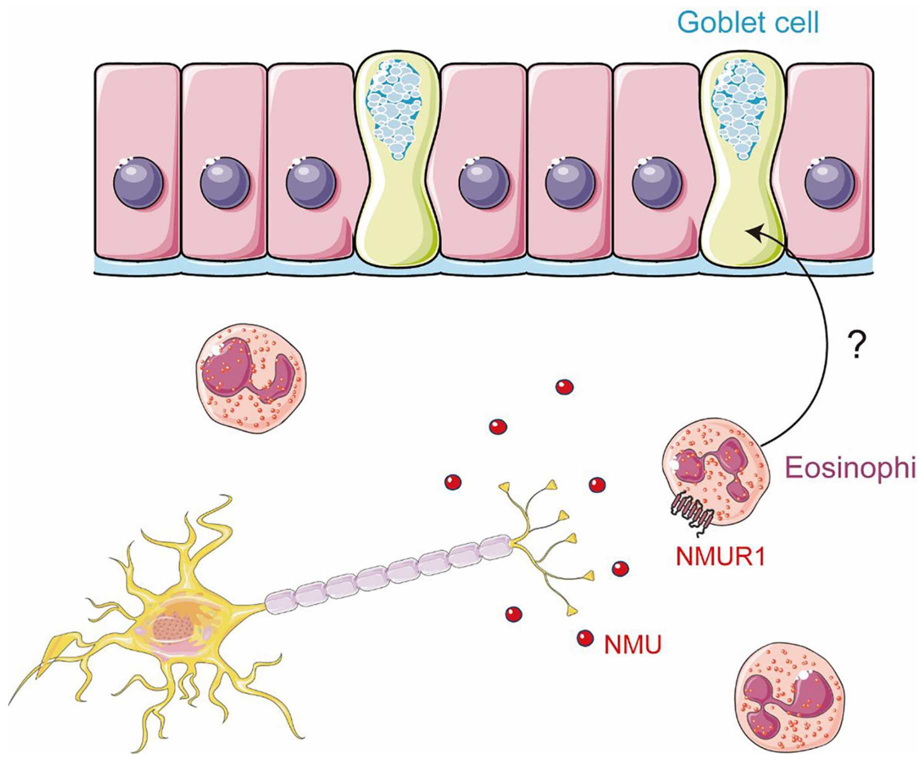

Fig. 1. Schematic of the role of neuromedin U in neuro-eosinophil crosstalk.

atopic dermatitis, and inflammatory bowel disease [13–15]. To investigate the role of NMUR1 in eosinophils in the context of inflammatory responses, Li and collaborators [11] used a variety of infection and disease models, including helminth infection, allergen stimulation, and dextran sulfate sodium-induced tissue damage and inflammation. Their observations suggest that the expression of NMUR1 on eosinophils is strictly regulated by the unique tissue microenvironment of the small intestine and is upregulated during inflammation. In addition, the authors generated a new eosinophil-specific Cre mouse strain and specifically depleted NMUR1 from eosinophils. With these new genetic tools, they found that NMUR1 regulated the degranulation process of the eosinophils, which was important to maintain small intestinal homeostasis, and the absence of NMUR1 reduced the number of eosinophils and increased the susceptibility of the organism to helminth infections.

In this new study, Li and colleagues [11] also found that NMUR1$^+$ eosinophils were able to promote the differentiation of goblet cells in the small intestine, which plays a crucial role in helminth expulsion [16]. To further explore the role of NMU-NMUR1 signaling in mucosal immunity of the small intestine, Li and colleagues [11] used electron microscopy and identified that eosinophils were in close proximity to neurons. Additionally, they adopted chemogenetic approaches—both activatory and inhibitory designer receptors exclusively activated by designer drugs (DREADDs)—to activate or inhibit NMU-expressing neurons and demonstrated that NMU could regulate goblet cell numbers and mucus secretion. Furthermore, to directly investigate the NMU-dependent eosinophil-goblet cell crosstalk, the authors employed an *in vitro* organoid culture system and discovered increased goblet cell differentiation when the intestinal organoids were cocultured with eosinophils plus NMU.

Based on these new findings, the authors proposed a model of neuro-immune-epithelium interaction in the small intestine, which relied on NMU-NMUR1 signaling (Fig. 1). This elegant model provides new insights into the complex interplay between the nervous system and immune system in type 2 immunity.

In the future, further in-depth research will be required to investigate the characteristics of neurons responsible for producing NMU and interacting with eosinophils. Additionally, it will be essential to delve into the molecular mechanisms underlying the regulation of goblet cells by NMUR1$^+$ eosinophils. This will pave the way for a more comprehensive understanding of the neuro-immune-epithelium interaction and potential treatments for conditions associated with small intestine mucosal immunity.

Conflict of interest

The authors declare that they have no conflict of interest.

Acknowledgments

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